USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: COMMON.CHECK.PD.12 Check Sink Capabilities Message
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
- Sink capabilities check update per the latest sink power rule in base spec - Modification to align to ECR AVS sink PDO deprecation in main PD group

Benefits as a result of the proposed changes:
Alignment to main specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None if DUT properly implemented

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

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Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

FROM:

COMMON.CHECK.PD.12: Check 1 Sink_Capabilities Message

Description: The Tester performs additional protocol checks on all *Sink_Capabilities* Messages sent by the UUT.
Check Applicability: The *Sink_Capabilities* Message
Perform the following checks on all *Sink_Capabilities* Messages: [COMMON.CHECK.PD.12#1]

1. The Tester runs the following field checks on the *Sink_Capabilities* Message:
    a. The Number of Data Objects field in the Message Header:
        i. Is at least one.
        ii. In PD3 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Snk_PDOs.
        iii. In PD2 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Snk_PDOs minus number of PPS PDOs (count of how many Snk_PDO_Supply_TypeN is set to PPS).
    b. For the first PDO, the Tester checks:
        i. B31…30 (Fixed Supply) set to 00b.
        ii. B29 (Dual-Role Power) set to 1b if VIF field PD_Port_Type is DRP, Provider/Consumer or Consumer/Provider, otherwise set to 0b.
        iii. B28: (Higher Capability) matches VIF field Higher_Capability_set
        iv. B27 - (Unconstrained_Power) matches VIF field Unconstrained_Power
        v. B26 (USB Communications Capable) matches VIF field USB_Comms_Capable
        vi. B25 (Dual-Role Data) set to 1b if VIF field DR_Swap_To_UFP_Supported or DR_Swap_To_DFP_Supported = YES, otherwise set to 0b if VIF field DR_Swap_To_UFP_Supported and DR_Swap_To_DFP_Supported = NO
        vii. B24…23
            1. PD3 mode (Fast Role Swap required USB Type-C Current) matches VIF field FR_Swap_Reqd_Type_C_Current_As_Initial_Source
            2. PD2 mode: set to 00b.
        viii. B22…20 (Reserved) set to 000b.
        ix. B19…10 (Voltage) set to 5V.
        x. B9…0 (Operational Current) matches VIF field SNK_PDO_OP_CURRENT1
    c. For each PDO, the Tester checks:
        i. The Tester checks consistency of Snk_PDO_Supply_TypeN
        ii. If Fixed Supply PDO:
            1. In PD2, B24…20 (Reserved) are set to zero
            2. In PD3, B22…20 (Reserved) are set to zero
            3. The Tester checks consistency of Snk_PDO_VoltageN
            4. The Tester checks consistency of Snk_PDO_ Op_CurrentN
        iii. If Variable Supply PDO:
            1. The Tester checks consistency of Snk_PDO_Max_VoltageN
            2. The Tester checks consistency of Snk_PDO_Min_VoltageN
            3. The Tester checks consistency of Snk_PDO_ Op_CurrentN
        iv. If Battery Supply PDO:
            1. The Tester checks consistency of Snk_PDO_Max_VoltageN
            2. The Tester checks consistency of Snk_PDO_Min_VoltageN
            3. The Tester checks consistency of Snk_PDO_ Op_PowerN
        v. If Programmable Power Supply APDO:
            1. B29…28 set to 00b.

2. B27…25 (Reserved) set to zero.
3. B16 (Reserved) set to zero.
4. B7 (Reserved) set to zero.
5. The Tester checks consistency of Snk_PDO_Max_VoltageN
6. The Tester checks consistency of Snk_PDO_Min_VoltageN
7. The Tester checks consistency of Snk_PDO_Max_CurrentN

d. The Tester compares all PDOs to VIF field PD_Power_as_Sink and check that they meet the requirements of the Power Rules
e. The Tester checks PDOs following the first one, are in the correct order: Fixed PDOs in increasing voltage sequence, Battery PDOs in increasing minimum voltage sequence, Variable PDOs in increasing minimum voltage sequence, and finally PPS PDOs in increasing maximum voltage sequence.
f. The Tester checks that no Fixed PDO has the same voltage as any other.
g. The Tester checks that no Variable PDO has the same voltage range as any other.
h. The Tester checks that no Battery PDO has the same voltage range as any other.
i. The Tester checks that no PPS PDO has the same voltage range as any other.

To:

COMMON.CHECK.PD.12: Check Sink_Capabilities Message

Description: The Tester performs additional protocol checks on all *Sink_Capabilities* Messages sent by the UUT.
Check Applicability: The *Sink_Capabilities* Message
Perform the following checks on all *Sink_Capabilities* Messages: [COMMON.CHECK.PD.12#1]

1. The Tester runs the following field checks on the *Sink_Capabilities* Message:
   a. The Number of Data Objects field in the Message Header:
      i. Is at least one.
      ii. In PD3 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Snk_PDOs.
      iii. In PD2 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Snk_PDOs minus number of PPS PDOs (count of how many Snk_PDO_Supply_TypeN is set to PPS).
   b. For the first PDO, the Tester checks:
      i. B31…30 (Fixed Supply) set to 00b.
      ii. B29 (Dual-Role Power) set to 1b if VIF field PD_Port_Type is DRP, Provider/Consumer or Consumer/Provider, otherwise set to 0b.
      iii. B28: (Higher Capability) matches VIF field Higher_Capability_set
      iv. B27 - (Unconstrained_Power) matches VIF field Unconstrained_Power
      v. B26 (USB Communications Capable) matches VIF field USB_Comms_Capable
      vi. B25 (Dual-Role Data) set to 1b if VIF field DR_Swap_To_UFP_Supported or DR_Swap_To_DFP_Supported = YES, otherwise set to 0b if VIF field DR_Swap_To_UFP_Supported and DR_Swap_To_DFP_Supported = NO
      vii. B24…23
         1. PD3 mode (Fast Role Swap required USB Type-C Current) matches VIF field FR_Swap_Reqd_Type_C_Current_As_Initial_Source
         2. PD2 mode: set to 00b.
      viii. B22…20 (Reserved) set to 000b.
      ix. B19…10 (Voltage) set to 5V.
      x. B9…0 (Maximum Current) is within allowed values (0-5A)
      xi. The tester checks V*I, where V= 5V and I = B9…0(Maximum/Operating Current)*10mA, does not exceed the max PDP in VIF field Snk_SPR_Maximum_PDP
   c. For each PDO, the Tester checks:
      i. The Tester checks consistency of Snk_PDO_Supply_TypeN
      ii. If Fixed Supply PDO:
         1. B31…30 set to 00b (Fixed Supply)

2. In PD2, B24…20 (Reserved) are set to zero
3. In PD3, B29…20 (Reserved) are set to zero
4. The Tester checks V*I, where V = B19…10 (Voltage)*50mV and I = B9…0 (Maximum/Operational Current)*10mA, does not exceed the max PDP in VIF field Snk_SPR_Maximum_PDP
5. The Tester checks B9…0 (Maximum/Operational Current) is within allowed current values (0-5A)

iii. If Variable Supply PDO:
1. B31…30 set to 10b (Variable)
2. The Tester checks V*I, where V = B19…10(Minimum Voltage)* 50mV and I = B9…0 (Maximum Current) x 10mA, does not exceed the max PDP in VIF field Snk_SPR_Maximum_PDP

iv. If Battery Supply PDO:
1. B31…30 set to 01b (Battery)
2. The Tester checks B9…0 (Maximum Power) does not exceed the max PDP indicated in VIF field Snk_SPR_Maximum_PDP

v. If Programmable Power Supply APDO:
1. B31…30 set to 11b (APDO)
2. B29…28 set to 00b.
3. B27…25 (Reserved) set to zero.
4. B16 (Reserved) set to zero.
5. B7 (Reserved) set to zero.
6. The Tester checks V*I, where V = B24…17(Maximum Voltage)*100mV and I = B6…0 (Maximum Current)*50mA, does not exceed the max PDP in VIF field Snk_SPR_Maximum_PDP

d. The Tester checks PDOs following the first one, are in the correct order: Fixed PDOs in increasing voltage sequence, Battery PDOs in increasing minimum voltage sequence, Variable PDOs in increasing minimum voltage sequence, and finally PPS PDOs in increasing maximum voltage sequence.
e. The Tester checks that no Fixed PDO has the same voltage as any other.
f. The Tester checks that no Variable PDO has the same voltage range as any other.
g. The Tester checks that no Battery PDO has the same voltage range as any other.
h. The Tester checks that no PPS APDO has the same voltage range as any other.
i. The Tester checks that there is no AVS APDO type in the list